SEMESTER III

6 Papers

Total 100 x 6 = 600 Marks

I. SKILL ENHANCEMENT COURSE SEC 1:

(Credits: Theory-02)

Marks : 100 (ESE: 3Hrs) =100 **Pass Marks Th ESE = 40**

Instruction to Question Setter for

End Semester Examination (ESE):

*There will be **objective type test** consisting of hundred questions of 1 mark each. Students are required to mark their answer on **OMR Sheet** provided by the University.*

ELEMENTARY COMPUTER APPLICATION SOFTWARES:

A Common Syllabus Prescribed by Ranchi University **Theory: 30 Lectures**

Objective of the Course

The objective of the course is to generate qualified manpower in the area of Information Technology (IT) and Graphic designing which will enable such person to work seamlessly at any Offices, whether Govt. or Private or for future entrepreneurs in the field of IT.

A. INTRODUCTION TO COMPUTER SYSTEM

Basic Computer Concept

Computer Appreciation - Characteristics of Computers, Input, Output, Storage units, CPU, Computer System. **(1 Lecture)**

Input and Output Devices

Input Devices - Keyboard, Mouse, joystick, Scanner, web cam,
Output Devices- Soft copy devices, monitors, projectors, speakers, Hard copy devices, Printers – Dot matrix, inkjet, laser, Plotters. **(4 lectures)**

Computer Memory and Processors

Memory hierarchy, Processor registers, Cache memory, Primary memory- RAM, ROM, Secondary storage devices, Magnetic tapes, Floppy disks, hard disks, Optical Drives- CD-ROM, DVD-ROM, CD-R, CD-RW, USB Flash drive, Mass storage devices: USB thumb drive. Managing disk Partitions, File System. Basic Processor Architecture, Processor speed, Types of processor.

(5 lectures)

Numbers Systems and Logic Gates

Decimal number system, Binary number system, Octal number system, Hexadecimal number system, Inter-conversion between the number systems. Basic Logic gates-AND, OR, NOT, Universal logic gates- NAND, NOR

(3 lectures)

Computer Software

Computer Software- Relationship between Hardware and Software, System Software, Application Software, Compiler, Names of some high level languages, Free domain software.

(2 Lectures)

Internet & its uses

History of Internet, WWW and Web Browsers: Web Browsing software, Surfing the Internet, Chatting on Internet, Basic of electronic mail, Using Emails, Document handling, Network definition, Common terminologies: LAN, WAN, MAN, Node, Host, Workstation, Bandwidth, Network Components: Severs, Clients, Communication Media. Wireless network

(3 Lectures)

Operating system-Windows

Operating system and basics of Windows, The User Interface, Using Mouse and Moving Icons on the screen, The My Computer Icon, The Recycle Bin, Status Bar, Start and Menu & Menu-selection, Running an Application, Windows Explorer Viewing of File, Folders and Directories, Creating and Renaming of files and folders, Opening and closing of different Windows, Windows Setting, Control Panels, Wall paper and Screen Savers, Setting the date and Sound, Concept of menu Using Help, Advanced Windows, Using right Button of the Mouse, Creating Short cuts, Basics of Window Setup, Notepad, Window Accessories

(2 Lectures)

B. MICROSOFT OFFICE 2007 AND LATEST VERSIONS

Word Processing

(2 Lectures)

B. MICROSOFT OFFICE 2007 AND LATEST VERSIONS

Word Processing

Word processing concepts: saving, closing, Opening an existing document, Selecting text, Editing text, Finding and replacing text, printing documents, Creating and Printing Merged Documents, Character and Paragraph Formatting, Page Design and Layout. Editing and Checking. Correcting spellings. Handling Graphics, Creating Tables and Charts, Document Templates and Wizards, Mail merge and Macros.

(3 Lectures)

Microsoft Excel (Spreadsheet)

Spreadsheet Concepts, Creating, Saving and Editing a Workbook, Inserting, Deleting Work Sheets, entering data in a cell / formula Copying and Moving from selected cells, handling operators in Formulae, Functions: Mathematical, Logical, statistical, text, financial, Date and Time functions, Using Function Wizard. Formatting a Worksheet: Formatting Cells changing data alignment, changing date, number, character or currency format, changing font, adding borders and colors, Printing worksheets, Charts and Graphs – Creating, Previewing, Modifying Charts. Integrating word processor, spread sheets, web pages. Pivot table, goal seek, Data filter and scenario manager

(4 Lectures)

Microsoft Power Point (Presentation Package)

Creating, Opening and Saving Presentations, Creating the Look of Your Presentation, Working in Different Views, Working with Slides, Adding and Formatting Text, Formatting Paragraphs, Drawing and Working with Objects, Adding Clip Art and other pictures, Designing Slide Shows, Running and Controlling a Slide Show, Printing Presentations. Creating photo album, Rehearse timing and record narration. Master slides. **(3 Lectures)**

Reference Books

- Nishit Mathur, Fundamentals of Computer , Aph publishing corporation(2010)
- Misty E. Vermaat,.Microsoft word 2013 1st Edition (2013).
- Satish Jain, M.Geeta, MS- Office 2010 Training Guide, BPB publication (2010)
- Joan Preppernau, Microsoft PowerPoint 2016 step by step, Microsoft press(2015)
- Douglas E Corner, The Internet Book 4th Edition, prentice –Hall(2009)
- Faithe wempen, word 2016 in depth 1st edition, que publishing(2015)
- Steven welkler, Office 2016 for beginners, Create Space Independent publishing Plateform (2016)

SKILL ENHANCEMENT LAB- SEC 1 LAB 30 Lectures

A. MS-WORD LAB ASSIGNMENT

1. Write down the following Paragraph OR any one provided by your teacher;

Without a doubt, the Internet is one of the most important inventions of modern times. The Internet is a global interconnected computer networks which allow each connected computer to share and exchange information with each other. The origins of the Internet can be traced to the creation of Advanced Research Projects Agency Network (ARPANET) as a network of computers under the auspices of the U.S. Department of Defense in 1969.

Apply following effects on The paragraph:

i. Paragraph **font-size** and **font-type** must be 12 Verdana.
ii. Paragraph **alignment** must be justified and double line spacing.
iii. **Highlight** the "(ARPANET)" with green color.
iv. Make the "Internet" keywords **Bold and Italic**.
v. Insert any "**WordArt**" and a **symbol** to your document.
vi. Insert a **clipart** to your document.
vii. Add following lines to your document:
Internet, Intranet, Extranet, URL, WWW, Networking, Protocols, HTTP, TCP/IP

2. Create a Table of following fields:

Name, Surname, Age, Gender, Job and apply the following effects

i. Insert 10 records
ii. Font size should be 12
iii. Title size should be 14
iv. Font type should be Times new Roman
v. Title color should be blue
vi. Text color should be black
vii. Table border should be 2

3. Write a letter on 'Road Safety' and send to 'Multiple Recipients' using mail merge.

4. Type the paragraph given below:

4. Type the paragraph given below:

Today, the Internet is a public, cooperative and self-sustaining facility accessible to hundreds of millions of people worldwide. Physically, the Internet uses a portion of the total resources of the currently existing public telecommunication networks. Technically, what distinguishes the Internet is its use of a set of protocols called TCP/IP (for Transmission Control Protocol/Internet Protocol). Two recent adaptations of Internet technology, the intranet and the extranet, also make use of the TCP/IP protocol. Today, the Internet is a public, cooperative and self-sustaining facility accessible to hundreds of millions of people worldwide. Physically, the Internet uses a portion of the total resources of the currently existing public telecommunication networks. Technically, what distinguishes the Internet is its use of a set of protocols called TCP/IP (for Transmission Control Protocol/ Internet Protocol). Two recent adaptations of Internet technology, the intranet and the extranet, also make use of the TCP/IP protocol.

Apply the following:

i. Change Internet into Internets at a time
ii. Heilight TCP/IP in red color
iii. Replace protocol into protocols
iv. Find the word "Public"

B. MICROSOFT EXCEL LAB ASSIGNMENT

Basic Formatting and Spreadsheet Manipulation

1. Add rows and columns to an existing spreadsheet
2. Reformat data (center, comma and currency styles, bold, text color)
3. Work with a simple formula (product) and function (sum)

Assignment

1. Create a workbook as shown below.
2. To enter new rows or columns, simply click on the row or column header to select the whole row or column. Then right click with the mouse and choose insert.
3. Add the new row for S Spade with the data that's shown below (between the original rows 7 and 8).
4. Add a column for gender and the data as shown below (between the original columns A and B). Enter the appropriate gender for yourself in the last row.

A	B	C	D
Name	**Male/Female**	**Genre**	**Number of Songs**
J Smith	F	Blues	50
B Doe	M	Country	110
S Spade	F	Country	200
F Zappa	M	Blues	1400
F Zappa	M	Alternative	2300
J Smith	F	Alternative	150
S Spade	F	Blues	1000
B Doe	M	Blues	75
yourname	M	Blues	800

5. Center the data in columns B and C. Do this by selecting the whole column and click the center icon on the ribbon.
6. Bold the data in row 1, the column headings (ensure that the data all remains visible within the column boundaries).
7. Change the font color for row 1 to Blue.
8. Change the format of the data in column D to comma style (no decimal places showing).There is an icon on the home tab that sets it to comma style easily.
9. Add two new column labels to the right of the current columns; **Unit Price** and **Total Cost**. (They will be in columns E and F.) These two columns of data should be currency type so that the dollar sign is shown. There is an icon to quickly format the selected column as currency type.
10. All tunes are $.99, so enter that value for all rows in Column E. You can copy quickly by using the **Auto Fill** handle and drag that amount down. When you over your mouse over the tiny square in the bottom right hand corner of the active cell, your mouse shape will become a skinny plus sign, and you can click and drag that cell to make a copy.

IV. CORE COURSE -C 5:

(Credits: Theory-04, Practicals-02)

Marks : 15 (MSE: 1Hr) + 60 (ESE: 3Hrs) =75 **Pass Marks: Th (MSE +ESE) = 30**

Instruction to Question Setter for
Mid Semester Examination (MSE):

*There will be **two** group of questions. **Group A is compulsory** and will contain five questions of **very short answer type** consisting of 1 mark each. **Group B will contain descriptive type** three questions of five marks each, out of which any two are to answer.*

End Semester Examination (ESE):
*There will be **two** group of questions. **Group A is compulsory** and will contain two questions. **Question No.1 will be very short answer type** consisting of ten questions of 1 mark each. **Question No.2 will be short answer type** of 5 marks. **Group B will contain descriptive type** five questions of fifteen marks each, out of which any three are to answer.*
***Note:** There may be subdivisions in each question asked in Theory Examinations.*

DATA STRUCTURES **Theory: 60 Lectures**

Arrays **(5 Lectures)**
Single and Multi-dimensional Arrays, Sparse Matrices (Array and Linked Representation)

Stacks **(5 Lectures)**
Implementing single / multiple stack/s in an Array; Prefix, Infix and Postfix expressions, Utility and conversion of these expressions from one to another; Applications of stack; Limitations of Array representation of stack

Linked Lists **(10 Lectures)**
Singly, Doubly and Circular Lists (Array and Linked representation); Normal and Circular representation of Stack in Lists; Self Organizing Lists; Skip Lists

Queues **(5 Lectures)**
Array and Linked representation of Queue, De-queue, Priority Queues

Recursion **(5 lectures)**
Developing Recursive Definition of Simple Problems and their implementation; Advantages and Limitations of Recursion; Understanding what goes behind Recursion (Internal Stack

Trees **(20 Lectures)**
Introduction to Tree as a data structure; Binary Trees (Insertion, Deletion , Recursive and Iterative Traversals on Binary Search Trees); Threaded Binary Trees (Insertion, Deletion, Traversals); Height-Balanced Trees (Various operations on AVL Trees).

Searching and Sorting **(5 Lectures)**
Linear Search, Binary Search, Comparison of Linear and Binary Search, Selection Sort, Insertion Sort, Insertion Sort, Shell Sort, Comparison of Sorting Techniques

Hashing **(5 Lectures)**
Introduction to Hashing, Deleting from Hash Table, Efficiency of Rehash Methods, Hash Table Reordering, Resolving collusion by Open Addressing, Coalesced Hashing, Separate Chaining, Dynamic and Extendible Hashing, Choosing a Hash Function, Perfect Hashing Function

Reference Books:

- Data Structures using C", by Y.Kanetkar.
- Data structure by R.B Patel
- Data structures using C by A.M Padma Reddy

Introduction to Tree as a data structure; Binary Trees (Insertion, Deletion , Recursive and Iterative Traversals on Binary Search Trees); Threaded Binary Trees (Insertion, Deletion, Traversals); Height-Balanced Trees (Various operations on AVL Trees).

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Hashing **(5 Lectures)**

Introduction to Hashing, Deleting from Hash Table, Efficiency of Rehash Methods, Hash Table Reordering, Resolving collusion by Open Addressing, Coalesced Hashing, Separate Chaining, Dynamic and Extendible Hashing, Choosing a Hash Function, Perfect Hashing Function

Reference Books:

- Data Structures using C", by Y.Kanetkar.
- Data structure by R.B Patel
- Data structures using C by A.M Padma Reddy

V. CORE COURSE -C 6: (Credits: Theory-04, Practicals-02)

Marks : 15 (MSE: 1Hr) + 60 (ESE: 3Hrs) =75	Pass Marks: Th (MSE +ESE) = 30

Instruction to Question Setter for

Mid Semester Examination (MSE):

*There will be **two** group of questions. **Group A is compulsory** and will contain five questions of **very short answer type** consisting of 1 mark each. **Group B will contain descriptive type** three questions of five marks each, out of which any two are to answer.*

End Semester Examination (ESE):

*There will be **two** group of questions. **Group A is compulsory** and will contain two questions. **Question No.1 will be very short answer type** consisting of ten questions of 1 mark each. **Question No.2 will be short answer type** of 5 marks. **Group B will contain descriptive type** five questions of fifteen marks each, out of which any three are to answer.*

***Note:** There may be subdivisions in each question asked in Theory Examinations.*

OPERATING SYSTEMS **Theory: 60 Lectures**

1. Introduction **(10 Lectures)**

Basic OS functions, resource abstraction, types of operating systems–multiprogramming systems, batch systems , time sharing systems; operating systems for personal computers & workstations, process control & real time systems.

2. Operating System Organization **(6 Lectures)**

Processor and user modes, kernels, system calls and system programs.

3. Process Management **(20 Lectures)**

System view of the process and resources, process abstraction, process hierarchy, threads, threading issues, thread libraries; Process Scheduling, non-pre-emptive and pre-emptive scheduling algorithms; concurrent and processes, critical section, semaphores, methods for inter-process communication; deadlocks.

Physical and virtual address space; memory allocation strategies -fixed and variable partitions, paging, segmentation, virtual memory

5. File and I/O Management **(10 Lectures)**

Directory structure, file operations, file allocation methods, device management.

6. Protection and Security **(4 Lectures)**

Policy mechanism, Authentication, Internal access Authorization.

Reference Books:

- A Silberschatz, P.B. Galvin, G. Gagne, Operating Systems Concepts, 8Edition,John Wiley Publications 2008.
- A.S. Tanenbaum, Modern Operating Systems, 3rd Edition, Pearson Education 2007.
- Gagne galvin, Operating Systems: 10th Edition .
- W. Stallings, Operating Systems, Internals & Design Principles , 5th Edition, Prentice Hall of India. 2008

5. File and I/O Management **(10 Lectures)**

Directory structure, file operations, file allocation methods, device management.

6. Protection and Security **(4 Lectures)**

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Reference Books:

- A Silberschatz, P.B. Galvin, G. Gagne, Operating Systems Concepts, 8Edition,John Wiley Publications 2008.
- A.S. Tanenbaum, Modern Operating Systems, 3rd Edition, Pearson Education 2007.
- Gagne galvin, Operating Systems: 10th Edition .
- W. Stallings, Operating Systems, Internals & Design Principles , 5th Edition, Prentice Hall of India. 2008

VI. CORE COURSE -C 7: (Credits: Theory-04, Practicals-01)

Marks : 15 (MSE: 1Hr) + 60 (ESE: 3Hrs) =75	Pass Marks: Th (MSE +ESE) = 30

Instruction to Question Setter for
Mid Semester Examination (MSE):

*There will be **two** group of questions. **Group A is compulsory** and will contain five questions of **very short answer type** consisting of 1 mark each. **Group B will contain descriptive type** three questions of five marks each, out of which any two are to answer.*

End Semester Examination (ESE):

*There will be **two** group of questions. **Group A is compulsory** and will contain two questions. **Question No.1 will be very short answer type** consisting of ten questions of 1 mark each. **Question No.2 will be short answer type** of 5 marks. **Group B will contain descriptive type** five questions of fifteen marks each, out of which any three are to answer.*

***Note:** There may be subdivisions in each question asked in Theory Examinations.*

COMPUTER NETWORKS THEORY **Theory: 60 Lectures**

1. Introduction to Computer Networks **(8 Lectures)**

Network definition; network topologies; network classifications; network protocol; layered network architecture; overview of OSI reference model; overview of TCP/IP protocol suite.

2. Data Communication Fundamentals and Techniques **(10 Lectures)**

Analog and digital signal; data-ratelimits; digital to digital line encoding schemes; pulse code modulation; parallel and serial transmission; digital to analog modulation-; multiplexing techniques-FDM, TDM; transmission media.

3. Networks Switching Techniques and Access mechanisms **(10 Lectures)**

Circuit switching; packets witching- connectionless datagram switching, connection-oriented virtual circuit switching; dial-up modems; digital subscriber line; cable TV for data transfer.

4. Data Link Layer Functions and Protocol **(10 Lectures)**

Error detection and error correction techniques ;data-link control- framing and flow control; error recovery protocols- stop and wait ARQ, go-back-n ARQ; Point to Point Protocol on Internet.

5. Multiple Access Protocol and Networks **(5 Lectures)**

CSMA/CD protocols; Ethernet LANS; connecting LAN and back-bone networks- repeaters, hubs, switches, bridges, router and gateways;

6. Networks Layer Functions and Protocols **(6 Lectures)**

Routing; routing algorithms; network layer protocol of Internet- IP protocol, Internet control protocols.

7. Transport Layer Functions and Protocols **(6 Lectures)**

Transport services- error and flow control, Connection establishment and release- three way handshake;

8. Overview of Application layer protocol **(5 Lectures)**

Overview of DNS protocol; overview of WWW & HTTP protocol.

Reference Books:

- B. A. Forouzan: Data Communications and Networking, Fourth edition, THM, 2007.
- S. Tanenbaum: Computer Networks, Fourth edition, PHI , 2002

Network Layer Functions and Protocols **(6 Lectures)**

Routing; routing algorithms; network layer protocol of Internet- IP protocol, Internet control protocols.

7. Transport Layer Functions and Protocols **(6 Lectures)**

Transport services- error and flow control, Connection establishment and release- three way handshake;

8. Overview of Application layer protocol **(5 Lectures)**

Overview of DNS protocol; overview of WWW & HTTP protocol.

Reference Books:

- B. A. Forouzan: Data Communications and Networking, Fourth edition, THM, 2007.
- S. Tanenbaum: Computer Networks, Fourth edition, PHI , 2002

COMPUTER PRACTICAL- C 5, C6 & C7 LAB

60 Lectures

Marks : Pr (ESE: 3Hrs)=75	Pass Marks: Pr (ESE) = 30

Instruction to Question Setter for

Practical Examination (ESE Pr)

*There will be **three** group of questions in Practical Examination of 3Hrs **Group A having questions from CORE PAPER 5(CC5)** will contain four questions , out of which any two are to be answered **Group B having questions from CORE PAPER 6(CC6)** will contain four questions , out of which any two are to be answered **Group C having questions from CORE PAPER 7(CC7)** will contain two questions , out of which any one is to be answered.*

***Lab**: Student have to Answer the given questions on Answer booklet and execute the answered programs/steps in computer with standard output.*

***Assignment**: The Assignment should be hand written in A4 size paper. First three pages (i.e. front page + acknowledgment + index) & Bibliography may be printout.*

***Marks Distribution**:*

LAB(Experiment + Answer script)	*= 45 marks*
Assignment	*=15 marks*
Viva-voce	*=15 marks*

COMPUTER PRACTICAL- C 5 LAB

1. Write a program to search an element from a list. Give user the option to perform Linear or Binary search. Use Template functions.
2. WAP using templates to sort a list of elements. Give user the option to perform sorting using Insertion sort, Bubble sort or Selection sort.
3. Implement Linked List using templates. Include functions for insertion, deletion and search of a number, reverse the list and concatenate two linked lists (include a function and also overload operator +).
4. Implement Doubly Linked List using templates. Include functions for insertion, deletion and search of a number, reverse the list.
5. Implement Circular Linked List using templates. Include functions for insertion, deletion and search of a number, reverse the list.
6. Perform Stack operations using Linked List implementation.
7. Perform Stack operations using Array implementation. Use Templates.
8. Perform Queues operations using Circular Array implementation. Use Templates.
9. Create and perform different operations on Double-ended Queues using Linked List implementation.
10. WAP to calculate factorial and to compute the factors of a given no. (i)using recursion, (ii) using iteration
11. WAP to display fibonacci series (i)using recursion, (ii) using iteration
12. WAP to calculate GCD of 2 number (i) with recursion (ii) without recursion
13. WAP to create a Binary Search Tree and include following operations in tree:
    a. Insertion (Recursive and Iterative Implementation)
    b. Deletion by copying
    c. Deletion by Merging
    d. Search a no. in BST
    e. Display its preorder, postorder and inorder traversals Recursively
    f. Display its preorder, postorder and inorder traversals Iteratively
    g. Display its level-by-level traversals
    h. Count the non-leaf nodes and leaf nodes
    i. Display height of tree
    j. Create a mirror image of tree
    k. Check whether two BSTs are equal or not

(ii) using iteration

11. WAP to display fibonacci series (i)using recursion, (ii) using iteration
12. WAP to calculate GCD of 2 number (i) with recursion (ii) without recursion
13. WAP to create a Binary Search Tree and include following operations in tree:
    a. Insertion (Recursive and Iterative Implementation)
    b. Deletion by copying
    c. Deletion by Merging
    d. Search a no. in BST
    e. Display its preorder, postorder and inorder traversals Recursively
    f. Display its preorder, postorder and inorder traversals Iteratively
    g. Display its level-by-level traversals
    h. Count the non-leaf nodes and leaf nodes
    i. Display height of tree
    j. Create a mirror image of tree
    k. Check whether two BSTs are equal or not

COMPUTER PRACTICAL- C 6 LAB

60 Lectures

1. Write a program (using *fork()* and/or *exec()* commands) where parent and child execute:
    a. same program, same code.
    b. same program, different code.-
    c. before terminating, the parent waits for the child to finish its task.
2. Write a program to report behaviour of Linux kernel including kernel version, CPU type and model. (CPU information)
3. Write a program to report behaviour of Linux kernel including information on configured memory, amount of free and used memory. (memory information)
4. Write a program to print file details including owner access permissions, file access time, where file name given as argument.
5. Write a program to copy files using system calls.
6. Write program to implement FCFS scheduling algorithm.
7. Write program to implement Round Robin scheduling algorithm.
8. Write program to implement SJF scheduling algorithm.
9. Write program to implement non-preemptive priority based scheduling algorithm.
10. Write program to implement preemptive priority based scheduling algorithm.
11. Write program to implement SRJF scheduling algorithm.
12. Write program to calculate sum of n numbers using *thread* library.
13. Write a program to implement first-fit, best-fit and worst-fit allocation strategies.

COMPUTER PRACTICAL-C 7 LAB

60 Lectures

SECTION I

1. Explain the following networking protocols with their syntax.
   Ping, ipstat, http, ftp, ip config, netstat, lpr.
2. Configure TCP
3. Configure Router
4. Configure Remote Machine
5. Internetworking devices:- NIC, Modems, Repeaters, Routers, Hubs, Bridges, Switches and Gateways.

SECTION II

1. Simulate Cyclic Redundancy Check (CRC) error detection algorithm for noisy channel.
2. Simulate and implement stop and wait protocol for noisy channel.
3. Simulate and implement go back n sliding window protocol.
4. Simulate and implement selective repeat sliding window protocol.
5. Simulate and implement distance vector routing algorithm
6. Simulate and implement Dijkstra algorithm for shortest path routing.

Total 100 x 1 = 100 Marks

III. GENERIC ELECTIVE (GE 3)

(Credits: Theory-04, Practicals-02)

Marks : 100 (ESE: 3Hrs) =100 **Pass Marks: Th ESE = 40**

Instruction to Question Setter for

End Semester Examination (ESE):

*There will be **two** group of questions. **Group A is compulsory** and will contain two questions. **Question No.1 will be very short answer type** consisting of ten questions of 1 mark each. **Question No.2 & 3 will be short answer type** of 5 marks. **Group B will contain descriptive type** six questions of twenty marks each, out of which any four are to answer.*

***Note:** There may be subdivisions in each question asked in Theory Examinations.*

CLIMATOLOGY

Theory: 60 Lectures

1. Atmospheric Composition and Structure – Variation with Altitude, Latitude and Season.
2. Insolation and Temperature – Factors and Distribution, Heat Budget, Temperature Inversion.
3. Atmospheric Pressure and Winds – Planetary Winds, Forces affecting Winds, General Circulation, Jet Streams.
4. Atmospheric Moisture – Evaporation, Humidity, Condensation, Fog and Clouds, Precipitation Types, Stability and Instability; Climatic Regions (Koppen)
5. Cyclones – Tropical Cyclones, Extra Tropical Cyclones, Monsoon - Origin and Mechanism.

Reading List

- Barry R. G. and Carleton A. M., 2001: *Synoptic and Dynamic Climatology*, Routledge, UK.
- Barry R. G. and Corley R. J., 1998: *Atmosphere, Weather and Climate*, Routledge, New York.
- Critchfield H. J., 1987: *General Climatology*, Prentice-Hall of India, New Delhi
- Lutgens F. K., Tarbuck E. J. and Tasa D., 2009: *The Atmosphere: An Introduction to Meteorology*,Prentice-Hall, Englewood Cliffs, New Jersey.
- Oliver J. E. and Hidore J. J., 2002: *Climatology: An Atmospheric Science*, Pearson Education, New Delhi.
- Trewartha G. T. and Horne L. H., 1980: *An Introduction to Climate*, McGraw-Hill.
- Gupta L S(2000): Jalvayu Vigyan, Hindi Madhyam Karyanvay Nidishalya, Delhi Vishwa Vidhyalaya,Delhi
- Lal, D S (2006): Jalvayu Vigyan, Prayag Pustak Bhavan, Allahabad
- Vatal, M (1986): Bhautik Bhugol, Central Book Depot, Allahabad
- Singh, S (2009): Jalvayu Vigyan, Prayag Pustak Bhawan, Allahabad

GEOGRAPHY PRACTICAL- GE3 LAB

60 Lectures

Marks : Pr (ESE: 3Hrs) =25 **Pass Marks: Pr (ESE) = 10**

Instruction to Question Setter for

End Semester Examination (ESE):

There will be one Practical Examination of 3Hrs duration. Evaluation of Practical Examination may be as per the following guidelines:

Experiment	*= 15 marks*
Practical record notebook	*= 5 marks*
Viva-voce	*= 5 marks*

PRACTICAL- STATISTICAL METHODS IN GEOGRAPHY

1. Tabulation and Descriptive Statistics: Frequencies (Deciles, Quartiles), Cross Tabulation,
2. Central Tendency (Mean, Median and Mode)
3. Dispersion (Standard Deviation, Variance and Coefficient of Variation) Scattered diagram, Histogram, Frequency Polygon & Frequency Curve

Class Record: Each student will submit a record containing five exercises:

1. Measures of central tendency and dispersion would be computed and interpreted for any two attributes.
2. Histograms and frequency curve would be prepared **and** interpreted for one or two variables.
3. Based on of the sample set and using two relevant attributes, a scatter and regression line would be

SEMESTER III **GENERIC ELECTIVE** **1 Paper**

Total 100 x 1 = 100 Marks

III. GENERIC ELECTIVE (GE 3)

(Credits: Theory-05, Tutorial-01)

Marks : 100 (ESE 3Hrs) =100	Pass Marks Th ESE = 40

Instruction to Question Setter for

End Semester Examination (ESE):

*There will be **two** group of questions. **Group A is compulsory** and will contain three questions. **Question No.1 will be very short answer type** consisting of ten questions of 1 mark each. **Question No.2& 3 will be short answer type** of 5 marks. **Group B will contain descriptive type** six questions of 20 marks each, out of which any four are to answer.*

***Note:** There may be subdivisions in each question asked in Theory Examinations.*

INDIAN ECONOMY

Theory: 75 Lectures; Tutorial:15 Lectures

Course Description

Using appropriate analytical frameworks, this course reviews major trends in economic indicators and policy debates in India in the post-Independence period with particular emphasis on paradigm shifts and turning points.

Course Outline

1. Economic Development since Independence

Major features of the economy at independence; growth and development under different policy regimes-goals, constraints, institutions and policy framework; an assessment of performance-sustainability and regional contrasts; structural change, savings and investment.

2. Population and Human Development

Demographic trends and issues; education; health and malnutrition

3. Growth and Distribution

Trends and policies in poverty; inequality and unemployment

4 .Agriculture (Policies and Performance)

Growth, Productivity, agrarian structure and technology, capital formation, pricing and procurement.

5. Industry (Policies and Performance)

Growth, productivity, diversification, small scale Industries, competition policy, foreign investment, Industrial Policy.